

Troubleshooting Azure Free Tier Errors

<https://gemini.google.com/app/d3a2bb354927abec>

User prompt: I want to solve a problem with a new free web app deployed with Azure App Service. Yesterday, I created a free web app using a new Azure App Service plan and a new resource as a free web app in Azure App Service (with the new Azure App Service plan mentioned) in the West US 3 region that was suggested by default by the Azure portal. After I created the new web app, I could not get the expected functionality of the web app because I received 503 or 403 error messages in my web browser SEVERAL times when accessing the web page, and it looked like it was related to "quota exceeded" in free Azure resources but I just published the app yesterday, and I don't believe that anyone use the new web app a lot. Can you help me diagnosing the origin of the problem and give me a solution?

Response: This is a classic AZ-104 scenario involving **App Service Plan SKU quotas, platform limitations, and app initialization spikes**.

Even without user traffic, a newly published application on the **Free (F1) tier** can easily trigger **HTTP 403 (Forbidden / Quota Exceeded)** or **HTTP 503 (Service Unavailable)** errors.

Origin of the Problem

- **Aggressive CPU Quotas on Free (F1):** The Free tier allows only **60 CPU minutes per day** across all apps sharing that plan. Heavy framework initialization (e.g., .NET, Java, or Node.js starting up, JIT compiling, or running database migrations) can consume 100% of a single core for a couple of minutes, rapidly depleting the daily quota.
- **403 vs. 503 Error Mechanism:**
 - **HTTP 403 (Forbidden / Quota Exceeded):** Returned by the Front-End Load Balancer when the **Daily CPU Quota (60 min)** or **Data Out Quotas (165 MB/day)** are exhausted. Azure stops serving traffic to protect shared host resources.
 - **HTTP 503 (Service Unavailable):** Happens when the app crashes during boot or runs out of physical memory (1 GB max). Free tier instances do not support **Always On**, so the process cold-starts whenever idle requests arrive, spiking CPU repeatedly.
- **Background Scanners & Health Probes:** Public Azure default domain names (*.azurewebsites.net) are constantly port-scanned by automated web bots within hours of creation, driving hidden CPU cycles.

Diagnostic Steps (AZ-104 Workflow)

Check the metric counters directly in the Azure Portal to confirm the exact root cause:

1. Navigate to your **App Service Plan** (not the App Service itself).
2. Go to **Settings > Quotas**. Check the status of **CPU Time**, **Memory**, and **Data Out**. If CPU Time is at 100%, the app will remain locked until the daily counter resets (every 24 hours UTC).

3. Go to **Monitoring > Metrics**:

- Add metric **CPU Time** (summed).
- Add metric **Http 4xx** and **Http 5xx**.

Solutions

Option 1: Upgrade to B1 (Recommended for AZ-104 Exam Scenarios) For production or active development, upgrade to at least the **Basic (B1)** tier:

- Removes daily CPU & Bandwidth quota enforcement.
- Enables **Always On** to eliminate cold-start spikes.
- Grants dedicated VM resources rather than shared host multi-tenancy.
- *How:* Go to **Scale Up (App Service Plan)** > Select **B1** (Dev / Test) > Click **Apply**.

Option 2: Work Around Free Tier Limits If you must keep the app on the **Free (F1)** tier:

- **Disable unnecessary startup jobs:** Ensure your application code is optimized to start up instantly without running heavy migrations, logging, or build scripts at runtime.
- **Scale Down / Restart:** Restart the App Service under **Overview > Restart** (only works if the hard daily CPU quota hasn't been completely hit).
- **Wait for Quota Reset:** If CPU time is fully exhausted, the platform unlocks the site automatically when the 24-hour UTC window resets.

AZ-104 Key Exam Takeaways

- **Free (F1) / Shared (D1):** Quotas applied (CPU, Memory, Bandwidth). *Always On* is **not** supported.
- **Basic (B1) & Above:** No quota limits on CPU time. Supports *Always On*, custom domains, and SSL.
- **Standard (S1) & Above:** Adds Auto-scaling, Staging Slots, and Automated Backups.

Sources

1. [How to Fix '503 Service Unavailable' Errors in Azure App Service](#)
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